

T $\equiv$ M Theory — Time Is Motion

A hypothesis on time and motion as the same principle

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Dedicated to my son and my wife, my guides in Tempo.

Abstract

This document presents an elemental idea:

Time and motion are not distinct phenomena but the same principle observed from two perspectives.

If the universe were to halt its movement—its expansion, vibration, or change—time itself would stop.

Cosmic expansion, material decay, and human perception of the passage of time would all be manifestations of a single reality: **universal motion**.

The proposal arises outside the academic sphere, from a personal intuition sustained over years, and seeks to express itself with clarity and simplicity, without unnecessary technicalities. Its purpose is to offer a direct and comprehensible reading of one of humanity's oldest questions:

What truly is time?

1. Genesis of the Idea

I do not come from the scientific or philosophical world.

I work with systems, processes, and logical structures, yet I always felt that something escaped technical definitions: time.

For years, the idea followed me with no intention to theorize it.

It simply appeared, insistently, every time I observed how everything moves, changes, and evolves.

One day in 2018, I wrote an email so as not to forget it:

"el tiempo es movimiento !!!"

And immediately saved the following draft:

"Si el universo frena su expansión, se frena el movimiento, el impulso que todo lo mueve, se frena el tiempo.

El tiempo es movimiento, es el desgaste de la materia en la velocidad a la cual viajamos en el universo."

That was the first conscious formulation.

Since then, the phrase has never left me.

That moment was more than an idea: it was a sudden certainty.

I felt that everything I had observed—from technical processes to natural cycles—fit into a single logic.

It was not a rational deduction but a complete perception, a clear intuition: **time and motion are one.**

From that moment I knew that, even without scientific training, I had seen something essential.

2. Central Proposition

“Time is motion.”

There is no time without motion, nor motion without time.

They are equivalent expressions of the same universal impulse of change.

It can be expressed symbolically as:

$$T \equiv M$$

where:

- **T** represents time, understood as perceived flow or duration.
- **M** represents motion, understood as universal variation of energy or state.

This equality is not mathematical but **ontological**: time does not *measure* motion; time *is* motion.

3. Conceptual Foundation

If each unit of time corresponds to a quantity of motion, it can be expressed as:

$$dt = k \, dM$$

where:

- **dt** = time interval
- **dM** = change in motion (energy, state, information)
- **k** = universal coherence constant that links motion variation to experienced duration

From this relation arise three principles:

1. If $dM = 0 \rightarrow dt = 0$: without motion, there is no time.
 2. Temporal flow is proportional to the total motion of the universe. The local temporal flow is modulated by relative motion and by the structure of the system.
 3. Cosmic expansion, atomic vibration, and consciousness are manifestations of the same fundamental movement.
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4. Conceptual Background

Various thinkers have approached this frontier, but none fully crossed it:

- **Aristotle** defined time as “the number of motion according to before and after”.
- **Newton** treated it as an absolute and independent variable.
- **Einstein** united time and space, showing that time deforms with velocity (motion).
- **Ernst Mach**, **Julian Barbour**, and **Carlo Rovelli** proposed that time might emerge from change.

All of them intuited the relationship, but none asserted its identity.

However, the **T $\equiv$ M hypothesis** proposes something more radical:
not that time depends on motion, but that it **is exactly the same principle**.

5. Implications

- **Cosmology:**
The expansion of the universe **is** the passage of time.
If expansion ceased, time would cease with it.
- **Relativity:**
Temporal dilation observed at high speeds confirms that time’s flow depends on relative motion.

- **Thermodynamics:**

The arrow of time aligns with the direction of motion toward increasing entropy.

- **Consciousness and Life:**

Human perception of time arises from internal motion: heartbeat, breathing, thought.

To live is to move; to exist is to last.

- **Limit State:**

A universe without motion would be a universe without time: a single motionless instant, without before or after.

6. Possible Objections

- **Circularity:**

The theory could be accused of being tautological.

Response: it does not redefine but identifies; it proposes that the distinction itself is a conceptual illusion.

- **Lack of mathematical formalization:**

There are still no experiments that directly measure the identity between time and motion.

Response: every theory begins as intuition. Its potential lies in coherence and simplicity.

- **Ambiguity of “motion”:**

The term is used broadly (energetic, quantum, informational).

Response: the breadth is deliberate; it seeks a unifying principle, not a limited model.

- **Empirical indistinguishability:**

One might argue that $T \equiv M$ is indistinguishable from models that separate time and motion.

Response: if both descriptions generate the same predictions, $T \equiv M$ offers conceptual parsimony without empirical loss.

7. Interdisciplinary Potential

- **Physics:** Redefine time as a variable derived from energetic motion.
- **Cosmology:** Reinterpret the expansion and end of the universe as motion–time cycles.
- **Biology:** View aging and vital rhythms as local manifestations of universal flow.
- **Neuroscience and AI:** Time perceived as a rhythm of processing or informational change.
- **Philosophy:** Unify “being” and “becoming”: what exists moves; what does not move, is not.

8. Verifiable Origin of the Idea (2018)



Figure 1. Capture of the original email (April 26, 2018), showing the first record of the T≡M idea.

That message marked the beginning of a thought that continued to evolve over the following years, until it crystallized into this formal proposal: the **T≡M Theory**.

9. Conclusion

No one has been able to explain time because time is not something distinct from motion.

The slowing of the universe's motion would be the slowing of time itself—the total pause. Matter would remain in a unique, motionless state, without before or after.

Time does not pass: it moves.
And when it stops moving, it ceases to exist.

10. Author's Note

I do not seek to impose a truth but to share a simple observation that seems to stand on its own:

Everything that exists moves.

Time, like life, is not a line: it is a continuous motion.
And this idea—born in an instant and following its own movement—is also proof of it.

Appendix I — Tempo and the verb *to tempe*

Origin

While developing this theory, it became necessary to name the fundamental action of the T≡M principle, which unifies time and motion.

In ordinary language, time *passes* and matter *moves*, but if both are the same phenomenon —as proposed here—, a single verb must express their unity.

Thus arises **to tempe**, and with it, the noun that contains it: **Tempo**.

The term **Tempo** derives from Latin *tempus*, whose Proto-Indo-European root **tem-* meant “to cut” or “to divide”, in the sense of **measuring flow or change**.

In classical usage, *tempus* also referred to the **rhythm or cadence of movement**, particularly in poetry and music, where it denoted the pulse of verse or sound.

Therefore, in its earliest sense, *tempus* already implied the **ordered rhythm of natural motion**.

The present concept of *Tempo* restores and extends that meaning: the principle unifying the passage of time and the displacement of motion in a single dynamic reality.

Definitions

Tempo: The fundamental entity unifying time and motion into one dynamic reality – the universal flow of being in transformation.

To tempe: The act of existing through unified time-motion; to participate in the universal flow of the Tempo.

Example: “The universe neither passes nor moves – it **tempes**.”

Distinction

- *To pass* expresses **time without matter** – the abstract flow.
- *To move* expresses **matter without time** – the physical displacement.
- *To tempe* unifies both: **the essential act of being, where motion occurs as time and time exists as motion.**

Reflection

Naming *to tempe* is not an invention but a recognition. The universe does not pass – **it tempes**.

Everything that exists does so in rhythm with its own **Tempo** – the pulse where time and motion remember they are the same.

This principle could be the basis for new ways of understanding the structure of the universe, but that will be another Tempo.

What passes without moving is imagined time.

What moves without passing is frozen matter.

What tempes, exists.

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